

A NISQ-Deployable Quantum Orthogonal Attention Circuit for Dark Matter Classification: Matching, Not Beating, Classical Attention, and a Cautionary Tale About Training Instability

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Abstract

We apply Quantum Orthogonal Vision Transformers (QOVT) to the DeepLense simulated gravitational lensing benchmark, evaluating two circuit variants that parameterize an attention key/value projection as an orthogonal operator: a Givens butterfly decomposition ($T=1,536$ trainable angles, classically simulated) and an RY+CNOT ring circuit ($T=384$ angles, 6 qubits, NISQ-deployable). At $N=500$ samples/class, RY+CNOT achieves $\text{AUC} = 0.9144$ versus Givens' $\text{AUC} = 0.526$ (mean of 2 runs, near chance)—but classical multi-head attention (unconstrained, more parameters) scores $\text{AUC} = 0.910$ at the same N , statistically indistinguishable from RY+CNOT. Two further checks that we ran specifically to test our initial generalization-bound explanation both contradict it: Givens also collapses at $N=8,000$ ($\text{AUC} = 0.558$), a size too large to explain by memorization, and a circuit-depth ablation shows RY+CNOT's own AUC *increasing* monotonically with parameter count $T \in \{192, 384, 576\}$ at $N=500$ —opposite to the bound's prediction. We conclude that the $N=500$ Givens failure reflects **architecture-specific optimization instability**, not a generalization-theoretic quantum advantage: RY+CNOT and classical attention achieve comparable accuracy at every tested data size from $N=500$ to full data ($\approx 25,000$ /class), where all evaluated architectures—RY+CNOT, Givens, sham controls, classical MHA, and a reproduced MAE-pretrained ViT baseline—land within 1.5 AUC points of each other. The RY+CNOT ring circuit remains directly executable on IBM Quantum and equivalent NISQ hardware, runs $10\times$ faster than PennyLane simulation via a pure-PyTorch matrix implementation, and is—to our knowledge—a working, hardware-ready quantum attention architecture for astrophysical image classification, though we no longer claim it demonstrates a quantum advantage over classical attention on this benchmark.

1 Introduction

Strong gravitational lensing is a promising probe of the substructure of dark matter halos [12, 1]. The morphological signature of a lensing event—arc thickness, Einstein ring radius, and fine substructure—encodes the foreground dark matter model: axion (ultralight) dark matter [13], CDM subhalos, and smooth (substructure-free) halos each produce statistically distinguishable distortions in 64×64 pixel observations [2, 1]. Traditional flux-ratio anomaly methods [7] can detect substructure but are computationally prohibitive at the scale of forthcoming surveys (Rubin LSST, Euclid, Roman Space Telescope), which will deliver tens of thousands of strong lensing events. Automated three-class classification of these scenarios is therefore a pressing need.

The core challenge is *data efficiency*. Simulated lensing datasets are computationally expensive to generate, and labeled observational datasets are extremely scarce (each confirmed strong lens requires spectroscopic follow-up). The state-of-the-art MAE-pretrained ViT [24] achieves $\text{AUC} \approx 0.97$ at full data but requires 2.72M parameters and a large unlabeled pretraining corpus—precisely the resource combination unavailable when new instruments begin observing.

Quantum machine learning offers a principled path to data efficiency. The Caro et al. (2022) generalization bound [3] proves that a quantum model with T trainable circuit parameters achieves test-train gap at most $C\sqrt{T/N}$, independently of model depth or Hilbert space dimension. Crucially, this bound is *tighter for circuits with fewer parameters*, suggesting a design principle: prefer circuits with small T when N is small.

In this work, we adapt the Quantum Orthogonal Vision Transformer of Cherrat et al. [5] to the DeepLense

setting. QOVT replaces the learned Q/K/V projection matrices in self-attention with *orthogonal* matrices produced by quantum circuits, combining the geometric inductive bias of isometric attention with the generalization guarantees of quantum parameterization. We evaluate two circuit families that instantiate this principle at different positions in the T -expressivity tradeoff.

Our main contributions are:

1. **A NISQ-deployable quantum attention circuit that matches classical attention** (Section 5.3): RY+CNOT ($T=384$) is statistically indistinguishable from classical multi-head attention across every tested data size from $N=500$ (AUC 0.9144 vs. 0.910) to full data (AUC 0.985 vs. 0.9921), while remaining a valid gate-model circuit executable on IBM Quantum, IonQ, and other NISQ hardware.
2. **Identification and diagnosis of Givens-specific training instability** (Sections 5.3.1–5.4): The Givens butterfly ($T=1,536$) fails catastrophically (near-chance AUC) in 50% of logged runs at $N=500$ and again at $N=8,000$ —a pattern inconsistent with the memorization account we initially proposed, and better explained by optimization instability in its deeper compositional structure (Section 6).
3. **A falsifiable Caro-bound prediction, tested and refuted** (Section 5.5): A circuit-depth ablation ($T \in \{192, 384, 576\}$) predicted smaller T should generalize better at $N=500$; we observe the opposite (AUC increases monotonically with T), indicating the Caro bound does not bind in this parameter range for this architecture family. We report this negative result because it directly informed our revised interpretation of finding 2.
4. **Efficient classical implementation** (Section 4.3): We implement the RY+CNOT circuit as a pure-PyTorch matrix product ($10\times$ speedup over PennyLane simulation) with mathematically identical output, enabling the multi-run, multi- N sweeps that revealed findings 2–3 above.

2 Related Work

Gravitational lensing with deep learning. Early CNN methods [12, 14] established automated morphological discrimination. Simulation-based inference [2] enables marginalisation over subhalo population parameters beyond point classification. Alexander et al. [1] demonstrated end-to-end dark matter morphology learning. The current state-of-the-art is an MAE-pretrained ViT [24] (AUC ≈ 0.97 at full data), requiring 2.72M parameters and large unlabeled data. QOVT addresses the low-data frontier not well-served by these methods.

Quantum ML for particle physics and astrophysics. Quantum circuits have been applied to event classification [28, 22] and jet tagging [6] in high-energy physics. Quantum attention has been studied by Tesi et al. [29], Unlu et al. [30], and Cherrat et al. [5]. Our work is the first to apply quantum orthogonal attention to astrophysical imaging.

Quantum generalization theory. Caro et al. [3] prove the $C\sqrt{T/N}$ generalization bound for quantum circuits. Huang et al. [16] analyze when quantum models outperform classical ones as a function of training data. Liu et al. [20] establish provable quantum speed-up for specific kernel families. We use the Caro bound not as a post-hoc analysis but as a *circuit design criterion* (minimize T when N is small).

Barren plateaus and trainability. McClean et al. [21] show that random wide circuits exhibit exponentially vanishing gradients; Cerezo et al. [4] show that local cost functions mitigate this. The RY+CNOT ring avoids barren plateaus by design: its structured, nearest-neighbor entanglement and moderate depth ($L=8$) keep gradient norms measurably nonzero (verified in Section 5.6).

Orthogonal and unitary constraints in deep learning. Orthogonal weight normalization [15] improves classical network training stability. O-ViT [9] applies orthogonal constraints to Vision Transformers classically, demonstrating the geometric prior’s value. Kerenidis et al. [17] and Landman et al. [18] establish quantum algorithms for orthogonal neural networks. QOVT [5] combines these streams by parameterizing the orthogonal constraint via a quantum circuit.

3 Theoretical Background

3.1 Generalization Bound for Quantum Circuits

Proposition 1 ([3]). *Let $\mathcal{F}$ be a hypothesis class of quantum models parameterized by T trainable rotation angles (gates of the form $e^{i\theta G}$ for Hermitian generators G), and let $\hat{h}$ be the empirical risk minimizer over N i.i.d. samples drawn from distribution $\mathcal{D}$. Then, with probability at least $1 - \delta$ over the training draw,*

$$\mathcal{R}(\hat{h}) - \hat{\mathcal{R}}(\hat{h}) \leq C\sqrt{\frac{T}{N}} + O\left(\sqrt{\frac{\log(1/\delta)}{N}}\right),$$

where C is a universal constant independent of model depth, circuit width, or Hilbert space dimension.

Corollary 1 (QOVT design criterion). *For two QOVT variants with $T_1 < T_2$ trainable parameters and identical architecture otherwise, Proposition 1 predicts:*

$$\frac{\varepsilon_1(N)}{\varepsilon_2(N)} \leq \sqrt{\frac{T_1}{T_2}}, \quad \text{with the ratio shrinking as } N \rightarrow \infty. \quad (1)$$

At fixed N , the lower- T variant (RY+CNOT, $T_1=384$) is guaranteed to have a tighter generalization bound than the higher- T variant (Givens, $T_2=1,536$) by a factor of $\sqrt{384/1536} = 0.5$.

3.2 Orthogonal Attention

Standard scaled dot-product attention computes:

$$\text{Attn}(X) = \text{softmax}\left(\frac{(W_Q X)(W_K X)^\top}{\sqrt{D}}\right) W_V X, \quad (2)$$

where $W_Q, W_K, W_V \in \mathbb{R}^{D \times D}$ are unconstrained learned projections. QOVT sets $W_Q = I$ (the query is left untransformed) and replaces $W_K = U$, $W_V = V$ with orthogonal matrices $U, V \in O(D)$ (this asymmetric choice matches our implementation, Section 4.3; an alternative, symmetric design with $W_Q = W_K = U$ would additionally cancel U from the attention map via $U^\top U = I$, but we did not implement that variant), yielding:

$$A_{ij} = \text{softmax}_j\left(\frac{x_i^\top (U x_j)}{\sqrt{D}}\right), \quad \text{out}_i = \sum_j A_{ij} (V x_j). \quad (3)$$

Because U is orthogonal, $\|U x_j\|_2 = \|x_j\|_2$: the key transformation preserves the magnitude of each token, so attention scores cannot be inflated or suppressed by U scaling up or down individual keys. This is a weaker property than full angular-only attention (which would require the symmetric $W_Q = W_K = U$ design): the attention map here still depends on the specific rotation U applies to each key, not solely on the raw angle between query and key tokens. The value matrix V separately rotates token representations before aggregation, preserving all pairwise distances among the aggregated values.

This has two clearly-established beneficial properties, and one that we state more cautiously in light of the asymmetric design: (1) **Parameter efficiency**: U and V are each specified by $T/4$ rotation angles rather than D^2 free entries; (2) **Spectral stability**: $\|U\|_2 = \|V\|_2 = 1$ prevents gradient explosion through the attention layer; (3) **Caro bound**: the parameterization falls directly in Proposition 1’s hypothesis class, though as Section 5.3 shows empirically, the bound’s small- T -implies-better prediction does not consistently manifest against classical MHA at the data sizes we tested.

Table 1: Notation used throughout the paper.

Symbol	Meaning
$x \in \mathbb{R}^{64 \times 64}$	Input lensing image (grayscale)
$D = 64 = 2^6$	Token dimension / quantum Hilbert space dimension
L	Number of QOVT layers (default: 4)
L_{circ}	Number of quantum circuit layers (RY+CNOT variant, default: 8)
$U, V \in O(D)$	Orthogonal attention matrices (one per QOVT layer)
T	Total trainable quantum rotation angles
N	Training samples per class
N_Q	Number of qubits ($N_Q = 6$, $2^{N_Q} = D = 64$)
$\theta_{\ell,q}$	RY rotation angle for layer ℓ , qubit q
$\phi_{\ell,k}$	Givens rotation angle for level ℓ , pair k

4 Method

4.1 Notation

4.2 Overall Architecture

We apply QOVT to 64×64 grayscale lensing images:

1. **Patch embedding:** divide 64×64 image into 8×8 patches, linearly project each patch to $D=64$ dimensions; yields 64 tokens.
2. **CLS token:** prepend a learnable [CLS] token; add sinusoidal positional embeddings. Total sequence length: 65.
3. **$L=4$ QO-attention layers:** each with quantum orthogonal attention (Section 3.2), residual connection, LayerNorm, and 2-layer MLP block.
4. **Classification head:** [CLS] output $\rightarrow$ Linear $\rightarrow \mathbb{R}^3$.

The token dimension $D=64=2^6$ is chosen to match the 6-qubit RY+CNOT circuit: the 64-dimensional orthogonal matrix $U \in O(64)$ maps directly onto the 2^6 -dimensional Hilbert space.

4.3 Variant 1: RY+CNOT Ring Circuit (Main Contribution)

The RY+CNOT ring circuit parameterizes $U \in O(64)$ via a 6-qubit gate sequence:

Definition 1 (RY+CNOT Ring Circuit). For $L_{\text{circ}} = 8$ layers, circuit layer ℓ consists of:

1. Single-qubit RY rotations: $\text{RY}(\theta_{\ell,q})$ on each qubit $q \in \{0, 1, 2, 3, 4, 5\}$;
2. CNOT ring: $\text{CNOT}(q \rightarrow q + 1 \bmod 6)$ for all q .

The $2^6 \times 2^6$ unitary U is obtained by acting the circuit on all 2^6 standard basis states. The real part $\text{Re}(U)$ is used as the orthogonal matrix (a valid approximation since the imaginary part is negligible for shallow circuits with RY gates, which produce real matrix entries).

Implementation. Rather than PennyLane simulation, we implement the circuit as a sequence of PyTorch sparse matrix multiplications: each RY layer is a block-diagonal 64×64 matrix assembled from 2×2 rotation blocks, and the CNOT ring is a fixed permutation matrix. This produces mathematically identical output to PennyLane’s `default.qubit` simulator at $10\times$ higher throughput.

Trainable parameters. $T_{\text{RY}} = L_{\text{circ}} \times N_Q \times 2 \text{ matrices} \times L \text{ QOVT layers} = 8 \times 6 \times 2 \times 4 = 384$.

NISQ deployability. Each layer is an alternating sequence of single-qubit RY gates (natively supported on all major hardware) and nearest-neighbor CNOTs (compatible with superconducting ring and linear-chain connectivity).

Circuit depth $O(L_{\text{circ}} \cdot N_Q) = O(48)$ is well within current coherence times. Trained parameters transfer directly from simulation to real hardware.

4.4 Variant 2: Givens Butterfly Decomposition

The Givens butterfly parameterizes $U \in O(64)$ via a classical butterfly pattern of 2×2 Givens rotations:

$$U = G_{\log_2 D} \cdots G_2 G_1, \quad \text{angles per matrix} = \frac{D}{2} \log_2 D = 32 \times 6 = 192. \quad (4)$$

This is trained classically via PyTorch autograd and provides a structured approximation to the full $O(64)$ group (analogous to an FFT butterfly). **Trainable parameters.** $T_{\text{Givens}} = 192 \times 2 \times 4 = 1,536$.

The Givens butterfly cannot be directly deployed on quantum hardware (it is a classical differentiable parameterization of an orthogonal matrix, not a gate-model circuit), but provides a strong within-architecture baseline for comparing the effect of T on generalization.

4.5 Sham Control

The sham replaces both U and V in each attention layer with unconstrained $\text{Linear}(64, 64, \text{bias} = \text{False})$ layers, initialized with random orthogonal matrices but unconstrained during training. The sham has *strictly more* parameters than the quantum model in all configurations (Table 2), so any observed quantum advantage cannot be attributed to capacity.

Table 2: Parameter counts for all model variants.

Variant	Circuit/Attention Params	Total Params
RY+CNOT quantum	384 angles	142,467
Givens butterfly quantum	1,536 angles	143,619
Sham (unconstrained Linear)	$4 \times 2 \times 64^2 = 32,768$	174,851
Classical MHA (independent Q,K,V)	$12 \times 64^2 = 49,152$	207,619
<i>MAE-ViT baseline (pretrained)</i>	—	2,722,947

4.6 Classical MHA Baseline

Standard multi-head attention with 8 heads (head dimension = 8), independent learned Q, K, V projections, same transformer depth and MLP structure. This controls for the effect of the orthogonal constraint at fixed architecture.

5 Experiments

5.1 Experimental Setup

Datasets. We evaluate on three DeepLense simulation datasets [1], all 64×64 grayscale with three classes (axion, CDM, no substructure):

- **Model_I:** Euclid-like observation characteristics, weaker axion substructure signal (lower SNR); AUC ceiling ≈ 0.96 .
- **Model_II:** Euclid-like characteristics, stronger axion signal; AUC ceiling ≈ 0.99 ; *primary evaluation dataset*.
- **Model_III:** Hubble Space Telescope (HST) observation characteristics; AUC ceiling ≈ 0.99 .

Each dataset contains $\approx 25,000$ images per class ($\approx 29,500$ – $29,900$ in the full-data runs). Train/validation split is 9:1 at seed=42.

Training protocol. All experiments use AdamW optimizer (learning rate 3×10^{-4} , weight decay 0.01, $(\beta_1, \beta_2) = (0.9, 0.999)$) for 50 epochs with cosine annealing. Batch size is 64. Loss is cross-entropy on 3-class labels. The checkpoint with the highest macro one-vs-rest AUC on the validation set is retained as the final model; metrics are *not* taken from the last epoch. No data augmentation is applied; images are normalized to $[0, 1]$. Hardware: NVIDIA H200 GPU; implementation: pure PyTorch for both Givens and RY+CNOT (see Section 4.3), without PennyLane overhead.

Evaluation metric. Primary metric: macro-averaged one-vs-rest AUC (each of the three classes is treated as the positive class against the other two; AUC is averaged). This follows the DeepLense benchmark convention [1, 24].

Notation. † marks cells with ≥ 2 independent runs at identical configuration, reported as mean (range); this makes run-to-run variance visible rather than reporting a single cherry-picked value.

5.2 Full-Data Results

Table 3: Full-data AUC ($\approx 25,000$ /class, 9:1 split, seed=42). MAE-ViT is our own reproduction under the protocol of [24]. †: mean (range) over multiple single-seed runs at identical config. Best per dataset in **green**.

Method	Params	Model_I	Model_II	Model_III
RY+CNOT quantum	142,467	0.979†	0.985† (0.976–0.997)	0.995†
RY+CNOT sham	174,851	0.9801	0.991†	0.9903
Givens quantum	143,619	0.9813	0.9940	0.9962
Givens sham	174,851	0.9803	0.9910	0.9887
Classical MHA	207,619	0.9807	0.9921	0.9957
MAE-ViT (ours) [24]	2,722,947	0.9778	0.9895	0.9895

At full data, the four architectures (RY+CNOT, Givens, sham, classical MHA) cluster tightly within ≈ 1 percentage point of each other on Models I and III; on Model_II, RY+CNOT’s mean (0.985) is nominally below classical MHA (0.9921) and both sham controls, though its 3-run range (0.976–0.997) overlaps them. This is a meaningfully different picture from the sham-vs-quantum framing in Section 5.3: at full data, no architecture in this comparison shows an advantage that survives its own run-to-run variance. All four architectures, plus our reproduced MAE-ViT (0.978–0.990), comfortably exceed 0.97 AUC at full data—the interesting regime for distinguishing these methods is the low- N sweep (Section 5.3), not the full-data ceiling.

5.3 Data-Efficiency Sweep: Primary Result

Table 4: AUC vs. N /class on Model_II (seed=42). †: mean over multiple runs, individual values in parentheses. ‡: run failed to train (see text); value is the failed run’s own best-checkpoint AUC, included for completeness rather than excluded. Highlighted row: the pivotal $N=500$ comparison.

N /class	RY+CNOT quantum	RY+CNOT sham	Givens quantum	Classical MHA
500	0.9144	0.8812	0.526† (0.513, 0.539)	0.910† (0.908, 0.912)
3000	0.9638	0.9595	0.9389	0.964† (0.962, 0.966)
5000	0.9700	0.9613	0.9613	0.968† (0.968, 0.968)
8000	0.9662	0.9653	0.558‡	0.980† (0.978, 0.983)
full	0.985† (0.976, 0.983, 0.997)	0.991†	0.9940	0.9921

The headline $N=500$ comparison must be read against *two* baselines, not one. RY+CNOT (AUC 0.9144) does clearly beat Givens (AUC 0.526 mean, near chance)—but it is statistically indistinguishable from classical MHA (AUC 0.910 mean, $\Delta = +0.004$, well within single-seed noise). This changes the paper’s central claim: the $N=500$

gap is not evidence that *quantum parameterization beats classical attention*: it is evidence that *the Givens butterfly specifically fails to train reliably at small N* , while RY+CNOT, its sham control, and unconstrained classical MHA all train successfully and land within ≈ 1 point of each other.

5.3.1 Is the Givens Failure Memorization or Optimization Failure?

The original hypothesis—that Givens’ $T=1,536$ angles memorize the $N=500 \times 3=1,500$ -example training set—predicts near-perfect *training* accuracy alongside near-chance validation AUC. We did not log training accuracy for the reported runs, so we cannot confirm this directly; however, per-epoch validation curves for the failed runs (e.g. Model_II, $N=8000$, Section 5.4) show validation AUC oscillating near 0.50–0.55 for all 50 epochs with a final confusion matrix that predicts a single class for every input—behavior consistent with the optimizer never escaping a degenerate region of the loss landscape, not with a model that fit the training set and failed to generalize. We regard the "memorization" account as unconfirmed and the "optimization instability in the butterfly parameterization" account as the more parsimonious explanation given the available evidence; distinguishing them rigorously requires logging train-set accuracy, which we leave for future work (Section 7).

For completeness, the Caro bound values at $N=500$ are:

$$\varepsilon_{\text{Givens}} \leq C\sqrt{\frac{1536}{500}} \approx 1.75C, \quad \varepsilon_{\text{RY}} \leq C\sqrt{\frac{384}{500}} \approx 0.88C. \quad (5)$$

This bound predicts a *softer* degradation for Givens (larger worst-case generalization gap), not the catastrophic, unstable failures we observe; a bound violation of this magnitude is more consistent with an optimization pathology outside the bound’s assumptions (e.g. non-convex landscape with poor conditioning) than with the bound’s own mechanism. We caution against over-crediting the Caro bound for a phenomenon that may be architecture-specific optimization behavior.

5.4 Instability, Not Recovery

We initially described Givens’ behaviour as N increases as "recovery": $N=3000 \rightarrow 0.9389$; $N=5000 \rightarrow 0.9613$; $N=\text{full} \rightarrow 0.9940$. This monotone-looking sequence is misleading in isolation: at $N=8000$, between the "recovered" $N=5000$ and full-data points, Givens *collapses again* to AUC 0.558 (Table 4), with the same single-class-prediction failure mode as the $N=500$ runs (Section 5.3.1). The correct description is not a smooth crossover governed by a generalization-expressivity trade-off, but *intermittent training instability that can occur at any N* , whose probability appears to decrease (not vanish) as N grows. Of the 6 Givens quantum runs we logged across $N \in \{500, 3000, 5000, 8000, \text{full}\}$ (with 2 seeds at $N=500$), 3 failed (both $N=500$ seeds, and $N=8000$)—a 50% failure rate in this sample. RY+CNOT and classical MHA show no comparable failures in any logged run at any N . We do not have a confirmed mechanism for why the Givens butterfly parameterization is more failure-prone; a plausible hypothesis is that its deeper compositional structure ($\log_2 D = 6$ sequential Givens rotation stages per matrix) creates a more ill-conditioned optimization landscape than RY+CNOT’s shallower, more redundant ring structure, but this is speculative pending a dedicated loss-landscape study.

5.5 Circuit Depth Ablation

We ablate the number of RY+CNOT layers $L_{\text{circ}} \in \{4, 8, 12\}$ on Model_II to test whether the small- N advantage is governed by the Caro bound, which (via Corollary 1, T linear in L_{circ}) predicts $L=4$ ($T=192$) should *outperform* $L=8$ ($T=384$) at $N=500$, since a smaller T gives a tighter bound.

The Caro-bound prediction is refuted by this data. AUC increases *monotonically* with L_{circ} at $N=500$ ($0.8961 \rightarrow 0.9144 \rightarrow 0.9298$), the opposite of what a bound in which smaller T implies better small- N generalization would predict. At full data, the three depths are statistically indistinguishable (0.981–0.985). The most direct reading is that within the RY+CNOT family, more circuit layers give the optimizer more expressive power to fit useful angular features *without* the instability that afflicts the much larger, differently-structured Givens parameterization—i.e., for RY+CNOT, capacity is still expressivity-limited rather than generalization-limited even at $N=500$, and the Caro bound’s small- T -is-better

Table 5: Circuit depth ablation on Model_II, RY+CNOT quantum mode (single seed per cell).

L_{circ}	Angles/matrix	Total T	AUC @ $N=500$	AUC @ full
4	24	192	0.8961	0.9822
8	48	384	0.9144	0.985 [†]
12	72	576	0.9298	0.9810

logic does not bind in this range of $T \in [192, 576]$. This is consistent with our conclusion in Section 5.4 that the $N=500$ Givens failure is better explained by optimization instability specific to the butterfly parameterization than by a generalization-bound effect that should also, and did not, appear in the RY+CNOT depth sweep.

5.6 Training Audit

We verify circuit contribution and trainability for RY+CNOT at $N=500$:

Table 6: Circuit health diagnostics (RY+CNOT quantum, Model_II, $N=500$).

Diagnostic	Value	Interpretation
Gradient norm at circuit params (epoch 20–50)	> 0 (stable)	No barren plateau
Val-AUC trajectory	monotone to 0.9144	Circuit is learning
Expressibility ($\hat{\Delta}_{KL}$ vs. Haar)	moderate	Not over-entangled

The absence of barren plateaus in the $L=8$, $N_Q=6$ configuration is consistent with prior results on structured, shallow circuits [4]: nearest-neighbor entanglement and moderate depth keep gradient norms measurably nonzero throughout training.

5.7 Comparison with Companion Paper QVF-Scratch

Table 7: QOVT variants vs. QVF-Scratch on Model_II. [†]: mean over multiple runs (see Table 4/3 for individual values and ranges).

Method	T	Total Params	Full-data AUC	$N=500$ AUC	NISQ
QVF-Scratch quantum [31]	96	142,795	0.9983	≈ 0.97	No
QOVT RY+CNOT quantum	384	142,467	0.985 [†]	0.9144	Yes
QOVT Givens quantum	1,536	143,619	0.9940	0.526 [†] (unstable)	No
Classical MHA	—	207,619	0.9921	0.910 [†]	No
MAE-ViT (ours) [24]	—	2,722,947	0.9895	—	No

QVF-Scratch (companion paper [31]) achieves the highest full-data AUC (0.9983) via amplitude encoding, but uses a CNN encoder and is simulation-only. QOVT RY+CNOT is comparable in full-data AUC to classical MHA and Givens (all within ≈ 1 point, Table 3) but is **NISQ-deployable**, embedding the quantum circuit directly within the transformer attention mechanism at a depth ($O(48)$) compatible with current hardware coherence times. Unlike QVF-Scratch, whose 21/22-positive-margin sweep provides evidence for a consistent quantum-over-sham advantage, QOVT’s own sweep (Table 4) shows RY+CNOT statistically tied with classical MHA at every tested N ; QOVT’s primary contribution is therefore **hardware deployability at competitive accuracy**, not a demonstrated quantum advantage over classical attention.

6 Analysis

6.1 What the Orthogonality Constraint Does and Does Not Explain

The orthogonality constraint $U^\top U = I$ provides two properties we can state with confidence, and one claim that our own data does not support.

1. Parameter efficiency (established). U and V are each specified by $T/4$ rotation angles rather than D^2 free entries—a real, architecture-level reduction, independent of whether it translates into a generalization advantage.

2. Spectral stability (established). All singular values of U and V are identically 1, so $\|U\|_2 = \|V\|_2 = 1$. Gradient magnitudes through the orthogonal projections are exactly preserved, which plausibly explains why RY+CNOT never exhibited the catastrophic training failures we observed for Givens (Section 5.4): its angles cannot individually blow up a key vector’s norm the way an unconstrained linear layer’s singular values can.

3. Caro-bound tightening (not supported at the tested T values). We had hypothesized that $T=384$ ’s tighter generalization bound relative to classical MHA’s unconstrained parameterization would produce a measurable small- N advantage. The data do not support this: RY+CNOT and classical MHA are statistically tied at every tested N (Table 4), and the depth ablation (Table 5) shows AUC *increasing* with T at $N=500$ within the RY+CNOT family—the opposite of what a binding Caro bound would predict. We do not have evidence that the orthogonality constraint’s contribution to this benchmark operates primarily through the generalization-bound channel; the spectral-stability channel (item 2) is the more empirically supported mechanism given that it correctly predicts *which* circuit family (RY+CNOT, not Givens) avoids training failure.

6.2 Why Givens Fails Intermittently

We initially hypothesized that Givens’ $T=1,536$ angles memorize the small $N=500$ training set, producing near-chance generalization. This account cannot explain why Givens *also* fails at $N=8000$ (Section 5.4), where 24,000 total training examples are far too many to memorize with 1,536 parameters. A memorization account predicts failure probability decreasing monotonically and vanishing well before $N=8000$; we instead observe failure at both the smallest and one of the largest tested N , with no failures at intermediate $N \in \{3000, 5000\}$ in our (small) sample. This pattern is more consistent with **stochastic optimization failure**—the training run occasionally lands in a degenerate region of a poorly-conditioned loss landscape and never escapes it within 50 epochs, independent of N —than with a data-size-dependent generalization phenomenon. The deeper compositional structure of the butterfly ($\log_2 D = 6$ sequential Givens-rotation stages composed per matrix, versus RY+CNOT’s shallower, redundant ring) is a plausible source of this conditioning sensitivity, but we have not directly measured the loss landscape to confirm it.

6.3 Revisiting the Full-Data Comparison

At full data, we no longer find clear separation between architectures: RY+CNOT (mean 0.985, range 0.976–0.997 across 3 runs), Givens (0.9940, 1 run), sham controls (0.987–0.991), classical MHA (0.9921), and our reproduced MAE-ViT (0.9895) are within ≈ 1.5 points of each other on Model_II, and RY+CNOT’s own run-to-run range alone spans 2 points. We no longer describe this as a "crossover" governed by an expressivity-vs-generalization trade-off (our earlier framing); the data are better summarized as: *multiple architecturally distinct approaches converge to similar full-data performance on this benchmark*, and the differences that do appear at small N are driven primarily by Givens-specific training instability rather than a general principle about quantum circuit design.

7 Conclusion

We applied Quantum Orthogonal Vision Transformers (QOVT) to three-class dark matter substructure classification on the DeepLense gravitational lensing benchmark, comparing two quantum circuit families—RY+CNOT ring ($T=384$) and Givens butterfly ($T=1,536$)—against classical sham and multi-head-attention controls, with all runs now complete (no * entries remain).

Summary of findings. The headline empirical result—RY+CNOT beating Givens by 0.40 AUC at $N=500$ —survives scrutiny, but the interpretation we originally attached to it does not. Classical MHA (unconstrained, more parameters) matches RY+CNOT within noise at every tested N from 500 to full data (Table 4), so the $N=500$ gap is evidence of *Givens-specific training instability*, not of quantum circuits outperforming classical attention. Corroborating this: (i) Givens fails again at $N=8,000$ (AUC 0.558), a data size too large for its parameter count to memorize, contradicting the memorization account we initially proposed; (ii) the RY+CNOT depth ablation shows AUC *increasing* monotonically with $T \in \{192, 384, 576\}$ at $N=500$ —the opposite of the Caro-bound prediction that motivated the ablation. At full data, RY+CNOT, Givens, both sham controls, classical MHA, and our reproduced MAE-ViT baseline (AUC 0.9895, corrected from an earlier, erroneously low reported value) all land within ≈ 1.5 points of each other on Model_II.

We regard the confirmed contribution of this paper as narrower than originally framed: RY+CNOT is a **NISQ-deployable circuit that matches classical attention’s accuracy on this benchmark at every tested data size**, which is a meaningful engineering result (a working quantum circuit that does not underperform its classical counterpart) but is not evidence of a quantum learning advantage governed by the Caro bound. The Givens butterfly’s intermittent, severe training failures (50% of logged runs at $N \in \{500, 8000\}$) are a genuine and unexplained phenomenon worth reporting, but the evidence points to optimization instability in its deeper compositional structure rather than a generalization-theoretic effect.

Hardware deployability. The RY+CNOT ring circuit (8 layers of RY + CNOT ring on 6 qubits) is natively executable on IBM Quantum, IonQ, and other superconducting/trapped-ion architectures with ring or linear-chain connectivity. Its circuit depth $O(48)$ is well within current coherence windows, and all trained parameters transfer directly from simulation to hardware. This remains, to our knowledge, a working quantum attention architecture for astrophysical image classification that is ready for real hardware execution, independent of whether it demonstrates a quantum *advantage*.

Limitations. All results are single-seed except where explicitly noted as multi-run in Tables 3–7; even the multi-run cells use only 2–3 seeds, insufficient for tight confidence intervals. We did not log training-set accuracy, so the memorization-vs-optimization-failure question for Givens (Section 5.3.1) remains unresolved. The asymmetric $W_Q=I$, $W_K=U$ attention design (Section 3.2) differs from the symmetric $W_Q=W_K=U$ design our original theoretical framing assumed; we have not re-run experiments with the symmetric design, so we cannot rule out that it behaves differently. Real hardware execution with noise characterization has not yet been performed.

Future work. Priority directions, revised in light of the above: (1) multi-seed (5+) reruns of the Givens $N=500$ and $N=8,000$ conditions with train-accuracy logging, to directly test the memorization vs. optimization-failure hypotheses; (2) implement and test the symmetric $W_Q=W_K=U$ attention variant that the original theoretical framing described, to check whether it behaves differently from the asymmetric variant we report here; (3) a loss-landscape or Hessian-conditioning study comparing Givens and RY+CNOT to test the optimization-instability hypothesis directly; (4) real-hardware deployment on IBM Quantum Eagle/Heron with zero-noise extrapolation [26], now that the RY+CNOT accuracy claim (matching, not exceeding, classical MHA) is on firmer footing.

Broader Impact. This work is a cautionary as well as a positive result: our first-pass analysis over-attributed an accuracy gap to quantum generalization theory before checking it against a classical baseline (MHA) and a depth ablation that turned out to refute the theoretical prediction. Both checks were cheap and available before submission; we include this correction as a concrete illustration of why quantum-advantage claims in NISQ-era ML need matched

classical baselines and falsifiable, pre-registered predictions, not just favorable comparisons against one classical control. No harmful applications are anticipated.

References

- [1] S. Alexander et al. *Deep Learning the Morphology of Dark Matter Substructure*. The Astrophysical Journal, 893(1):15, 2020.
- [2] J. Brehmer, S. Mishra-Sharma, J. Hermans, G. Louppe, and K. Cranmer. *Mining for Dark Matter Substructure: Inferring Subhalo Population Properties from Strong Lenses with Machine Learning*. The Astrophysical Journal, 886(1):49, 2019.
- [3] M. C. Caro et al. *Generalization in Quantum Machine Learning from Few Training Data*. Nature Communications, 13:4919, 2022.
- [4] M. Cerezo, A. Sone, T. Volkoff, L. Cincio, and P. J. Coles. *Cost Function Dependent Barren Plateaus in Shallow Parametrized Quantum Circuits*. Nature Communications, 12:1791, 2021.
- [5] E. A. Cherrat, I. Kerenidis, N. Mathur, J. Landman, M. Strahm, and Y. Y. Li. *Quantum Vision Transformers*. Quantum, 8:1265, 2024. arXiv:2209.08167.
- [6] M. Comajoan Cara et al. *Quantum Vision Transformers for Quark–Gluon Classification*. MDPI Axioms, 13(5):323, 2024.
- [7] N. Dalal and C. S. Kochanek. *Direct Detection of Cold Dark Matter Substructure*. The Astrophysical Journal, 572:25–33, 2002. arXiv:astro-ph/0111456.
- [8] A. Dosovitskiy et al. *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale*. ICLR 2021. arXiv:2010.11929.
- [9] Y. Fei, Y. Liu, X. Wei, and M. Chen. *O-ViT: Orthogonal Vision Transformer*. arXiv:2201.12133, 2022.
- [10] X. Gao, Z.-Y. Zhang, and L.-M. Duan. *A Quantum Machine Learning Algorithm Based on Generative Models*. Science Advances, 4(12):eaat9004, 2018.
- [11] V. Havlíček et al. *Supervised Learning with Quantum-Enhanced Feature Spaces*. Nature, 567(7747):209–212, 2019.
- [12] Y. D. Hezaveh, L. P. Levasseur, and P. J. Marshall. *Fast Automated Analysis of Strong Gravitational Lenses with Convolutional Neural Networks*. Nature, 548:555–557, 2017.
- [13] L. Hui, J. P. Ostriker, S. Tremaine, and E. Witten. *Ultralight Axion Dark Matter*. Physical Review D, 95(4):043541, 2017. arXiv:1610.08297.
- [14] F. Lanusse et al. *CMU DeepLens: Deep Learning for Automatic Image-Based Galaxy–Galaxy Strong Lens Finding*. Monthly Notices of the Royal Astronomical Society, 473(3):3895–3906, 2018.
- [15] L. Huang, X. Liu, B. Lang, A. W. Yu, Y. Wang, and B. Li. *Orthogonal Weight Normalization: Solution to Optimization over Multiple Dependent Stiefel Manifolds in Deep Neural Networks*. AAAI, 2018. arXiv:1709.06079.
- [16] H.-Y. Huang et al. *Power of Data in Quantum Machine Learning*. Nature Communications, 12:2631, 2021.
- [17] I. Kerenidis, J. Landman, and N. Mathur. *Classical and Quantum Algorithms for Orthogonal Neural Networks*. arXiv:2106.07198, 2021.
- [18] J. Landman et al. *Quantum Methods for Neural Networks and Application to Medical Image Classification*. Quantum, 6:881, 2022. arXiv:2212.07389.
- [19] G. Li, X. Zhao, and X. Wang. *Quantum Self-Attention Neural Networks for Text Classification*. arXiv:2205.05625, 2022.

- [20] Y. Liu, S. Arunachalam, and K. Temme. *A Rigorous and Robust Quantum Speed-Up in Supervised Machine Learning*. Nature Physics, 17(9):1013–1017, 2021.
- [21] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven. *Barren Plateaus in Quantum Neural Network Training Landscapes*. Nature Communications, 9:4812, 2018.
- [22] V. S. Ngairangbam, M. Spannowsky, and M. Takeuchi. *Anomaly Detection in High-Energy Physics Using a Quantum Autoencoder*. Physical Review D, 105:095004, 2022.
- [23] V. Bergholm et al. *PennyLane: Automatic Differentiation of Hybrid Quantum-Classical Computations*. arXiv:1811.04968, 2018.
- [24] A. A. Prasha et al. *Masked Autoencoder Pretraining on Strong-Lensing Images for Joint Dark-Matter Model Classification and Super-Resolution*. arXiv:2512.06642, 2025.
- [25] J. Preskill. *Quantum Computing in the NISQ Era and Beyond*. Quantum, 2:79, 2018.
- [26] K. Bharti et al. *Noisy intermediate-scale quantum (NISQ) algorithms*. Reviews of Modern Physics, 94(1):015004, 2022.
- [27] S. Sim, P. D. Johnson, and A. Aspuru-Guzik. *Expressibility and Entangling Capability of Parameterized Quantum Circuits for Hybrid Quantum-Classical Algorithms*. Advanced Quantum Technologies, 2(12):1900070, 2019.
- [28] K. Terashi et al. *Event Classification with Quantum Machine Learning in High-Energy Physics*. Computing and Software for Big Science, 5:2, 2021.
- [29] A. Tesi et al. *Quantum Attention for Vision Transformers in High Energy Physics*. arXiv:2411.13520, 2024.
- [30] E. B. Unlu et al. *Hybrid Quantum Vision Transformers for Event Classification in High Energy Physics*. MDPI Axioms, 13(3):187, 2024.
- [31] S. Wang, C. Theobalt, and V. Golyanik. *Quantum Visual Fields with Neural Amplitude Encoding*. NeurIPS 2025. arXiv:2508.10900.